

QUALITY OF LIFE IN NON-INSULIN-DEPENDENT DIABETES AND A COMPARISON WITH INSULIN-DEPENDENT DIABETES

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Abstract—The reported effects of diabetes on quality of life have been assessed in two groups of attenders at out-patient clinics: 1. One hundred and twenty-one non-insulin-dependent diabetic patients randomly allocated to diet, tablet or ultralente insulin therapy; 2. Fifty-seven patients with insulin-dependent diabetes consecutively attending an out-patient clinic. The overall picture for those with non-insulin-dependent diabetes was of relatively little disruption to most areas of life, but 27% reported considerable loss of enjoyment and reduction in social life. High fasting plasma glucose was significantly associated with fatigue and leisure difficulties. The type of therapy, tablet, diet or insulin, made little difference to psychological, social or attitude variables. Those with insulin dependent diabetes showed similar psychological morbidity, but described a rather different pattern of social consequences with more effects on work and less on leisure.

INTRODUCTION

Most people with diabetes appear to be remarkably little affected by the illness and do not report significant psychological symptoms. A minority suffer considerable emotional distress together with fatigue, irritability and poor concentration [1]. Diabetes can also have considerable consequences for the quality of every-day life, with limitation of work, social life, family relations, sex and leisure interests [2]. In insulin-dependent diabetes (IDDM), these problems are said to be associated with poor glycaemic control as well as with major physical complications such as severe retinopathy, painful neuropathy and renal failure [1]. Psychiatric disorder is believed to be especially common in those with 'brittle diabetes' [3].

Less is known about the psychological and social aspects of non-insulin-dependent diabetes (NIDDM) and of possible associations with the severity of diabetes, type of treatment, obesity or other medical or social variables [4, 5, 6]. The alternative treatments, diet, tablets or insulin, might be expected to have different effects on patients' way of life.

We have examined the problems occurring in daily life in a random sample of NIDDM patients participating in the U.K. Prospective Diabetes Study [7] who attended their research outpatient clinic in Oxford over a 4-month period. They constitute a group of relatively recent onset without major complications. A comparison group of IDDM patients was drawn from consecutive attenders at the same diabetic physician's NHS out-patient clinic.

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The aims of the study were:

- (1) To describe patients' reports of the effects of NIDDM on their everyday lives.
- (2) To examine possible associations between reported problems and type of treatment and other variables.
- (3) To compare the reports of patients with NIDDM with a sample of patients with IDDM.

METHOD

Subjects with non-insulin-dependent diabetes

The U.K. Prospective Diabetes Study (UKPDS) recruits recently diagnosed patients with NIDDM aged 25–65 inclusive. The majority of patients referred to participating hospitals since 1980 are recruited into the study, excluding those with medical complications, occupations precluding randomisation, severe physical illness or an inadequate understanding of English. They are invited to participate in a study of diet, oral (Chlorpropamide, Glibenclamide or Metformin) and insulin (Ultralente) treatment [7]. If after three months on diet the fasting plasma glucose remains > 6 mmol/l (108 mg dl), patients are randomly allocated to the different therapies, the aim of each being to maintain the fasting plasma glucose to < 6 mmol/l. The mean age at diagnosis of the Oxford UKPDS patients is 52 years, and their mean weight is 120% above ideal body weight.

A random sample of 146 names was drawn from the weekly lists of attenders over a four-month period. Of these, 121 subjects were interviewed, 69 men and 52 women, with a mean age of 55.5 years (men: 55.1; women: 56.0), age range 32–71 years; a further 15 attended the clinic but were missed or could not wait, and 10 did not attend. These 25 subjects did not differ from those interviewed on any social or medical factor with the exception of duration of diabetes.

The final numbers interviewed in the three randomised treatment groups were: tablets, 63 (52%) (23 randomised to Chlorpropamide, 28 to Glibenclamide, and 12 subjects were prescribed Metformin), insulin, 35 (29%) and diet, 23 (19%). This is a slightly higher proportion on tablets than in the whole Oxford UKPDS clinic.

Subjects with insulin-dependent diabetes

The sample of IDDM patients was recruited from two diabetic clinics providing care for the Oxford area. Fifty-seven IDDM subjects, 32 men and 25 women, attending the clinic during the research period were interviewed; a further 27 were not interviewed because they failed to keep their clinic appointment, or in a few cases were missed at the clinic or could not wait for interview.

The assessments were carried out in the clinics, and consisted of a social difficulty questionnaire, and a mood questionnaire. Medical information was extracted from research records and case-notes.

Social Difficulty Questionnaire. This was adapted for the study on the basis of pilot interviews with diabetic subjects from a questionnaire used in previous studies of heart disease [8]. The questionnaire covered difficulty in aspects of leisure, work and family life over the last three months, and in complying with treatment. Subjects were interviewed and asked to rate the extent to which diabetes had limited various activities 'or made them more difficult or awkward or less enjoyable' on a 4-point scale from 'not affected at all' to 'greatly affected', and to describe any difficulties in detail. Overall measures of leisure, work and family difficulty were obtained by summing the ratings in each of these areas (expressed as a proportion of applicable activities). Subjects were also asked to indicate their agreement, on 5-point attitude scales, to four attitude statements about diabetes. Three of these were taken from a study by Bradley *et al.* [9].

Mental state. The Profile of Mood States (POMS) mood scale was used to assess mental state over the last week [10]. The five subtests, tension, depression, fatigue, anger and lack of vigour, were felt to be particularly appropriate for diabetes.

Clinical data were obtained from the case notes and from the extensive data already available in the UKPDS study.

Statistical analysis

The analysis was done on the SPSSX computer package, using χ^2 as the test of significance for categorical data. Non-parametric tests (Mann-Whitney U test and Kruskal-Wallis one-way analysis of variance) were used for the POMS scores analysis since these scores are not normally distributed.

RESULTS

Table I shows the demographic and medical characteristics of the two samples. Patients with NIDDM were older and had all suffered from diabetes for 10 years or less. Relatively few had developed major physical complications.

Quality of life in NIDDM

Effects on everyday activities

Most subjects reported relatively little disruption of their lives, other than minor inconveniences (rated as slight), such as having to remember to carry snacks or sweets (see Table II). Although a third reported either no effects or only slight effects, across all aspects of everyday life, a fifth reported that six or more aspects of everyday activities were affected.

Subjects were asked to describe the effects in more detail. Not everyone did so, while others gave several reasons (Table III). Tiredness was seen as a major cause of restrictions in work and leisure. The need to be careful about diet and alcohol was often blamed for a loss of pleasure in social life such as having an evening out with friends and visiting restaurants, pubs and clubs. Many had completely stopped these activities.

Mental state

Most subjects reported few psychological symptoms, but in a sizeable minority scores on the POMS mood scales for tension, depression, fatigue, anger and lack of vigour were high. Complaints of fatigue and lack of vigour were significantly more common in those who were more than 20% overweight. Mood scores were strongly associated with our measures of reported social difficulty, especially leisure and family (Table IV).

Variables associated with effects on quality of life in NIDDM

Demographic and medical factors

Sex. There were few differences in social consequences for men and women. More women than men felt that the preparation of meals had been made more difficult ($p < 0.05$). Rather more men than women felt their social life (38% compared with 21%) and their sex lives (33% compared with 15%; $p < 0.05$) had deteriorated. Women reported more tension ($p < 0.01$) and fatigue ($p < 0.05$).

Age. Leisure difficulties were related to age. Those aged in their fifties reported more effects on leisure than those under 50 or 65 or over ($p < 0.05$).

Medical problems. Few of these patients had major complications of their diabetes although a considerable number had minor ones. There was no association between the presence of complications (41%) and mental state. Adverse effects on quality of life were not associated with age of onset (under the age of 40 or over) or duration of diabetes.

Those who were obese on entry to the UKPDS study ($>20\%$ overweight) described more leisure and family difficulties ($p < 0.05$).

TABLE I.—CHARACTERISTICS OF 121 PATIENTS WITH NIDDM AND 57 PATIENTS WITH IDDM

	NIDDM		IDDM	
	N	%	N	%
Sex				
Men	69	(57)	32	(56)
Women	52	(43)	25	(44)
Mean age	55.5		34.9	
Age at onset				
Over 40	101	(84)		
25–40	20	(17)	27	(47)
< 25	—	—	30	(52)
Marital status				
Married	101	(84)	31	(54)
Single	8	(7)	20	(35)
Widowed	5	(4)	1	(2)
Divorced/separated	7	(6)	5	(9)
Employment status				
Work (full time)	64	(53)	41	(72)
Work (part time)	14	(12)	3	(5)
Sick	1	(1)	2	(4)
Not working	15	(12)	6	(11)
Retired	27	(22)	5	(9)
Duration of diabetes				
1–2 yr	42	(35)	10	(18)
3–5 yr	39	(32)	10	(18)
6–10 yr	40	(33)	10	(18)
11–20 yr	—	—	17	(28)
> 20 yr	—	—	10	(18)
Weight > 20% above target	59	(49)	Not available	
Fasting glucose				
< 7 mmol/l	40	(33)	20	(54)
7–8 mmol/l	43	(36)	10	(27)
9+ mmol/l	38	(31)	7	(19)
nk			20	
Treatment at interview				
Diet only	16	(13)		
Chlorpropamide	19	(16)		
Glibenclamide	26	(22)		
Metformin	13	(11)		
Insulin × 1	30	(25)	7	(12)
× 2	17	(14)	42	(74)
× 3			3	(5)
Other			5	(9)
Diet problem in notes				
Nil	79	(65)	40	(76)
Minor	22	(18)	7	(13)
Major	20	(17)	6	(11)
nk			4	
Therapy problem in notes				
Nil	97	(80)	24	(45)
Minor	10	(8)	14	(26)
Major	14	(12)	15	(28)
nk			4	
Psychosocial problems in notes				
Nil	95	(79)	41	(77)
Minor	17	(14)	8	(15)
Major	9	(7)	4	(8)
nk			4	

(Continued on facing page).

TABLE I.—*Continued*

	NIDDM		IDDM	
	N	%	N	%
Complications				
Retinopathy	8	(7)	15	(28)
Cataracts	5	(4)	1	(2)
Neuropathy	24	(20)	5	(9)
Claudication	1	(1)	1	(2)
Amputation	1	(1)	—	—
Ischaemic heart disease	—	—	2	(4)
Hypoglycaemic episodes*				
Nil			36	(68)
Transient			9	(17)
Temporary			6	(11)
Incapacitating			2	(4)
nk			4	

*Not applicable to NIDDM.

TABLE II.—EFFECTS OF DIABETES ON EVERYDAY LIFE

	N*	NIDDM			N*	IDDM		
		Slight %	Moderate %	Marked %		Slight %	Moderate %	Marked %
Leisure								
Shopping	115	15	6	5	54	17	2	—
Walking > 2 miles	115	15	3	13	51	14	4	8
Household jobs	119	16	9	6	52	21	8	—
Cooking	96	7	5	6	42	12	2	—
Social life	111	18	15	16	57	40	7	2
Holidays/outings	108	11	6	9	54	30	11	3
Eating out	102	27	11	17	50	34	6	8
Physical activities	66	12	8	6	40	23	15	5
Work								
Work routine	79	19	5	3	48	17	4	6
Hours worked	79	7	3	0	48	15	2	2
Travelling-work	78	9	3	0	48	10	2	4
Meal breaks	75	12	3	1	47	23	4	2
Concentration	79	15	3	1	48	31	4	—
Type of job (career)	79	10	1	5	48	9	12	12
Needing time off	79	13	4	1	48	29	6	—
Family								
Sex life	100	5	9	18	45	9	7	9
Worry for spouse	101	35	19	15	38	24	32	11
Worry for child < 16	27	11	7	7	12	17	25	—
Worry for others								
in home	55	22	9	9	25	16	32	12
Diet change spouse	101	26	21	9	38	32	24	—
Diet change < 16	27	37	15	11	12	42	8	—
Diet change-others	55	15	13	4	25	32	16	8
Disagreement with family	114	12	4	1	54	20	4	4

*N, number for whom activity applicable.

Control of diabetes

Fasting glucose. Only 3% of those with fasting glucose levels below 7 mmol/l had high overall leisure difficulty scores compared with 27% of those with higher glucose levels ($p < 0.01$).

TABLE III.—PROPORTIONS REPORTING REASONS FOR EFFECTS ON PARTICULAR ACTIVITIES

Effect	NIDDM %	IDDM %	Significance of χ^2 *
Going out socially	(<i>N</i> = 43)	(<i>N</i> = 19)	
Reduced/stopped	74	16	0.001
Less enjoyable	33	6	0.05
Planning	7	68	0.001
Eating out	(<i>N</i> = 31)	(<i>N</i> = 10)	
Reduced/stopped	64	0	0.01
Less enjoyable	23	10	NS
Planning	16	60	0.001
Worry finding right food	24	10	NS
Household tasks	(<i>N</i> = 28)	(<i>N</i> = 11)	
Tiredness	49	27	0.1
Shopping	(<i>N</i> = 25)	(<i>N</i> = 7)	
Tiredness	40	0	NS
Planning	0	71	0.01
Walking 2 miles+	(<i>N</i> = 26)	(<i>N</i> = 10)	
Tiredness	46	20	NS
Physical limitation	38	40	NS
Work routine	(<i>N</i> = 19)	(<i>N</i> = 10)	
Tiredness	73	20	0.05
Planning	5	60	0.05
Type of work possible	(<i>N</i> = 17)	(<i>N</i> = 23)	
Heavy or shift work difficult	41	65	NS
Job too stressful/tiring/had changed job/stopped work	52	9	0.01
Career or promotion impaired	0	43	0.01

*Test of significance: χ^2 with Yates Correction; df 1.

TABLE IV.—NIDDM: MEAN SCORES ON POMS BY OVERALL LEISURE, WORK AND FAMILY DIFFICULTY

	Low difficulty	High difficulty†	<i>p</i> -value* <
Leisure	(<i>N</i> = 96)	(<i>N</i> = 23)	
Tension	4.45	7.30	0.01
Depression	2.20	5.36	0.001
Fatigue	3.76	7.74	0.001
Anger	2.68	4.61	0.05
Lack of vigour	11.00	12.09	NS
Work	(<i>N</i> = 75)	(<i>N</i> = 9)	
Tension	4.36	8.44	NS
Depression	2.49	6.89	0.05
Fatigue	4.33	7.67	NS
Anger	2.90	5.67	0.05
Lack of vigour	10.85	14.22	0.01
Family	(<i>N</i> = 91)	(<i>N</i> = 22)	
Tension	4.31	7.41	0.01
Depression	1.95	6.27	0.001
Fatigue	3.78	7.18	0.001
Anger	2.53	5.05	0.001
Lack of vigour	10.78	12.31	0.05

†High difficulty = summed ratings of moderate or marked difficulty as percentage of applicable activities: highest 20% approx.

*Test of significance = Mann-Whitney U test.

High fasting plasma glucose was significantly associated with effects on walking, shopping and cooking. An association with fatigue remained significant after controlling for obesity. Poor control was not significantly associated with the other mood scales.

Glycosylated haemoglobin. We considered three groups of subjects: good (<8.5), medium ($8.5-10.0$) and poor (>10) control. The results were similar to those described for fasting glucose, those with poor and medium control of the diabetes, as measured by HbA_{1c} levels, generally reporting more adverse effects on their lives. These were significant for shopping, cooking, sexual relationships and worry for spouse, and for overall leisure difficulty (all $p < 0.05$) (data were not available for 35 patients).

Type of treatment

Randomised treatment. Overall there were few significant associations between any of the psychological, social or attitude variables and type of treatment at randomisation. However, those randomised to insulin were more likely to report difficulties (slight, moderate or marked) at work, and those on diet less likely to do so (Table V).

Current treatment. Most subjects were still on their randomised treatment at interview. However, 16 had been reallocated to improve control: 12 to insulin (9 from tablets, 3 from diet) and 4 to tablets (from diet).

The 28 insulin treated patients working at the time of the interview were more likely to describe difficulty with hours worked, concentration at work and the type of work they felt they could do (all $p < 0.05$), problems which were attributed to tiredness, anxiety or the difficulties of fitting long, late, or irregular hours into an insulin routine.

TABLE V.—RANDOMISATION GROUPS FOR TREATMENT OF NIDDM: REPORTED EFFECTS ON ACTIVITIES

Activities	N	Randomisation group				Significance of χ^2
		Tablets %	Insulin %	Diet %	Total %	
Leisure (moderate, severe)						
Shopping	115	10	9	17	11	
Walking > 2 miles	115	17	16	13	16	
Household jobs	119	11	15	26	15	
Cooking	96	6	13	24	12	
Social life	111	36	27	27	32	
Holidays	108	9	23	18	15	
Eating out	102	27	31	24	28	
Work (slight, moderate, severe)						
Daily routine	79	27	40	7	27	(0.1)
Hours worked	79	9	20	0	10	(0.1)
Concentration	79	22	20	7	19	
Family (moderate, severe)						
Sex-life	100	23	31	32	27	
Worry for spouse	101	28	41	36	34	
Disagreement	114	3	6	9	5	
High overall difficulty						
Leisure	121	16	20	26	19	0.05
Work	86	18	44	7	23	
Family	115	15	27	18	19	
Any	121	16	23	22	19	

TABLE VI.—TREATMENT AT INTERVIEW. NIDDM PATIENTS REPORTING EFFECTS OF DIABETES ON WORK

	Diet (<i>N</i> = 12) <i>N</i> %	Tablets (<i>N</i> = 39) <i>N</i> %	Insulin × 1 daily (<i>N</i> = 13) <i>N</i> %	Insulin × 2 daily (<i>N</i> = 15) <i>N</i> %	Significance of χ^2 *
Daily work routine	1 (8)	8 (21)	3 (23)	9 (60)	0.05
Hours worked	—	2 (5)	1 (8)	5 (33)	0.05
Travelling	1 (8)	3 (8)	1 (8)	4 (27)	NS
Meal breaks	1 (5)	6 (17)	2 (15)	3 (21)	NS
Concentration	—	6 (15)	2 (15)	7 (47)	0.05
Type of job possible	—	4 (10)	2 (15)	7 (47)	0.01
Needing time off	1 (8)	8 (21)	3 (23)	2 (13)	NS
Two or more areas affected		11 (28)	5 (39)	10 (67)	0.05

*Significance tests for Insulin × 1 vs Insulin × 2: χ^2 with Yates correction; df 1.

The fifteen patients who had been changed from the initial once-daily regimen to twice-daily insulin reported more effects on work than those treated with once-daily ultralente (Table VI). In contrast, there were no differences between those treated with ultralente insulin once a day and those treated with tablets.

There were differences in attitudes to the three types of treatment ($p < 0.05$). Only 7% of those on tablets reported difficulty with their treatment, compared with 31% on diet and 38% on insulin. Some of the insulin group strongly disliked having to inject themselves and four depended on relatives to do it for them. More of those on twice-daily than once-daily insulin agreed that they would enjoy life more if they were not diabetic ($p < 0.01$).

Quality of life in IDDM

Reported effects on social activities for the IDDM group are summarised in Table II. The patterns were rather different from those reported by the group with NIDDM. Significantly fewer subjects mentioned moderate or great effects on at least one leisure area (33% compared with 51% of the NIDDM group), but more rated some aspect of work as affected (46% compared with 23% of the NIDDM group).

Many IDDM subjects found the need to plan their activities around their insulin regimens irritating, frustrating, and worrying. Some complained of a loss of spontaneity, especially in their social lives, being unable to do things on the spur of the moment with their friends.

There were few sex differences in the IDDM group, but three times as many men as women reported worry for their spouse (60% compared with 20%; $p < 0.05$).

Mental state. Scores on the POMS mood scales were similar to those described from NIDDM. However, there was much less evidence in the IDDM group of associations between mood and effects on quality of life: only lack of vigour with overall leisure difficulty and fatigue with overall family difficulty were significant ($p < 0.05$).

Variables associated with quality of life

Control. We had limited data on fasting plasma glucose and HBA1c levels in the IDDM subjects (37 and 45 respectively of the 57 subjects). In contrast to NIDDM, neither measure was associated with fatigue or overall leisure difficulty. High levels of both were associated with effects on physical activities ($p < 0.05$).

Medical factors. Subjects who were liable to hypoglycaemic episodes (9 men and 8 women) reported more worry for their spouses, especially coping at night. They were more tense and lacking in vigour (all $p < 0.05$). They reported more difficulty in activities involving travelling or eating out, and were more likely to feel that 'diabetes affects my whole life-style' ($p < 0.001$).

Fifteen of the IDDM group (6 men and 9 women) had some retinopathy, but this was not associated with more reported social or psychological difficulties. Psychosocial factors were not related to duration of diabetes.

DISCUSSION

Non-insulin-dependent diabetes

We describe the first detailed picture of how people with NIDDM report that their condition had affected their lives. The overall picture for subjects with diabetes of relatively recent onset and few major complications is encouraging. The planning, self-care and attention to diet that diabetes requires were usually seen as having slight rather than devastating effects on activities. Even so, a sizeable number felt their lives were restricted by the illness. They complained of a loss of enjoyment of previously valued social pursuits, especially those involving social eating and drinking. The finding that nearly three in ten said they had cut down considerably on their social life or stopped going out altogether, is a matter of concern. Our sample of patients had relatively few physical complications of diabetes, and an older group with a longer duration of illness might be expected to report greater effects on quality of life.

Tiredness was a major complaint of these patients, and the clear association between high fasting glucose and fatigue, even when controlling for obesity, is interesting. Other workers have described individuals with IDDM who report clear associations between their mood and blood glucose but have not found consistent general associations [11, 12]. In the present study, too, high glucose levels were not associated with greater tension or depression.

Norms for medical and psychiatric populations are not available for the POMS mood scales, but in other studies of medical groups POMS scores have been found to correlate highly with the Present State Examination Index of Definitions [8, 13]. As a very rough indication of psychiatric disorder in the present study, nearly one-fifth of the NIDDM women had anxiety and tension scores above the mean POMS scores for a group of women awaiting hysterectomy who were diagnosed as PSE psychiatric cases [12]. This would suggest a sizeable minority with considerable psychological distress. Mental state and quality of life were closely associated and there were also associations between impaired leisure and poor control of diabetes.

There were encouragingly few differences in effects on quality of life between patients randomised to the different treatments. This would suggest that concern about psychological and social consequences should not play a significant part in considering the eventual conclusions of the UKPDS. However, those currently treated with diet or insulin found these less acceptable treatments, and more of those on a twice-daily insulin regime reported difficulties at work, mainly rated as slight.

Those treated with one injection a day reported no more social difficulties than those on tablets. Diet alone was unpopular because it was more demanding and worrying than the less strict dieting required of the other treatment conditions.

Comparison of NIDDM and IDDM

The patterns of social effects attributed to insulin-dependent diabetes [14], were rather different from those described by subjects with NIDDM, fewer effects on leisure and more on work, confined to fewer areas, and very different explanations for adverse effects. The NIDDM group reported considerable tiredness, loss of enjoyment and reduction in leisure pursuits, and some restrictions on life at work. The IDDM group had not, on the whole, reduced their activities, and effects on work were less to do with tiredness and stress and more to do with career and promotion losses; the main consequence for them was the need to plan regular meals rather than tiredness. Thus only 7% said that diabetes had resulted in reduced social activities as compared with 27% of the NIDDM group. Significantly fewer of the IDDM group agreed that 'diabetes is a serious illness' (58% compared with 73% in the NIDDM group; $p < 0.05$), and very few (only 5% compared with 18% of the NIDDM subjects) agreed that 'it is impossible to control my diabetes well and live in a way acceptable to me' ($p < 0.05$).

We conclude that the IDDM group were concerned to fit management of their diabetes around everyday activities which they were not prepared to forego, whereas the NIDDM group had modified their lives to fit in with their diabetes. Such findings may reflect differences between the diabetic disorders or perhaps more likely the differences in age of onset, age at the time of the survey and duration of illness. Previous psychological research has often been on very mixed diabetic populations. Future research will need to consider more precisely defined patient groups in which age, type of diabetes, medical complications and demographic characteristics are explicit.

These very different patterns of adjustment may explain our finding of virtually no association between social difficulties and psychological state in the IDDM group but of a strong association in NIDDM. Anxiety, depression and fatigue in the NIDDM subjects may be directly due to the disruptive consequences for long-established patterns of work, leisure and family life of a late-onset condition. However, we should be cautious about cause and effect; social withdrawal and fatigue can also be symptoms of depression and of physical illness.

Management

Physicians in diabetic medicine accept that good treatment of diabetes involves not only education but a variety of interventions to minimise the psychosocial impact of the illness and to improve compliance with medical advice. Most previous treatment has been concerned with IDDM [15, 16], but the results reported in this paper show that in NIDDM there is a similar need to minimise psychological distress and unnecessary impairment of quality of life. Intervention should concentrate on the minority of patients who report emotional distress and social difficulties. Successful treatment of psychosocial complaints might also have benefits for self-care and quality of control of the diabetes.

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